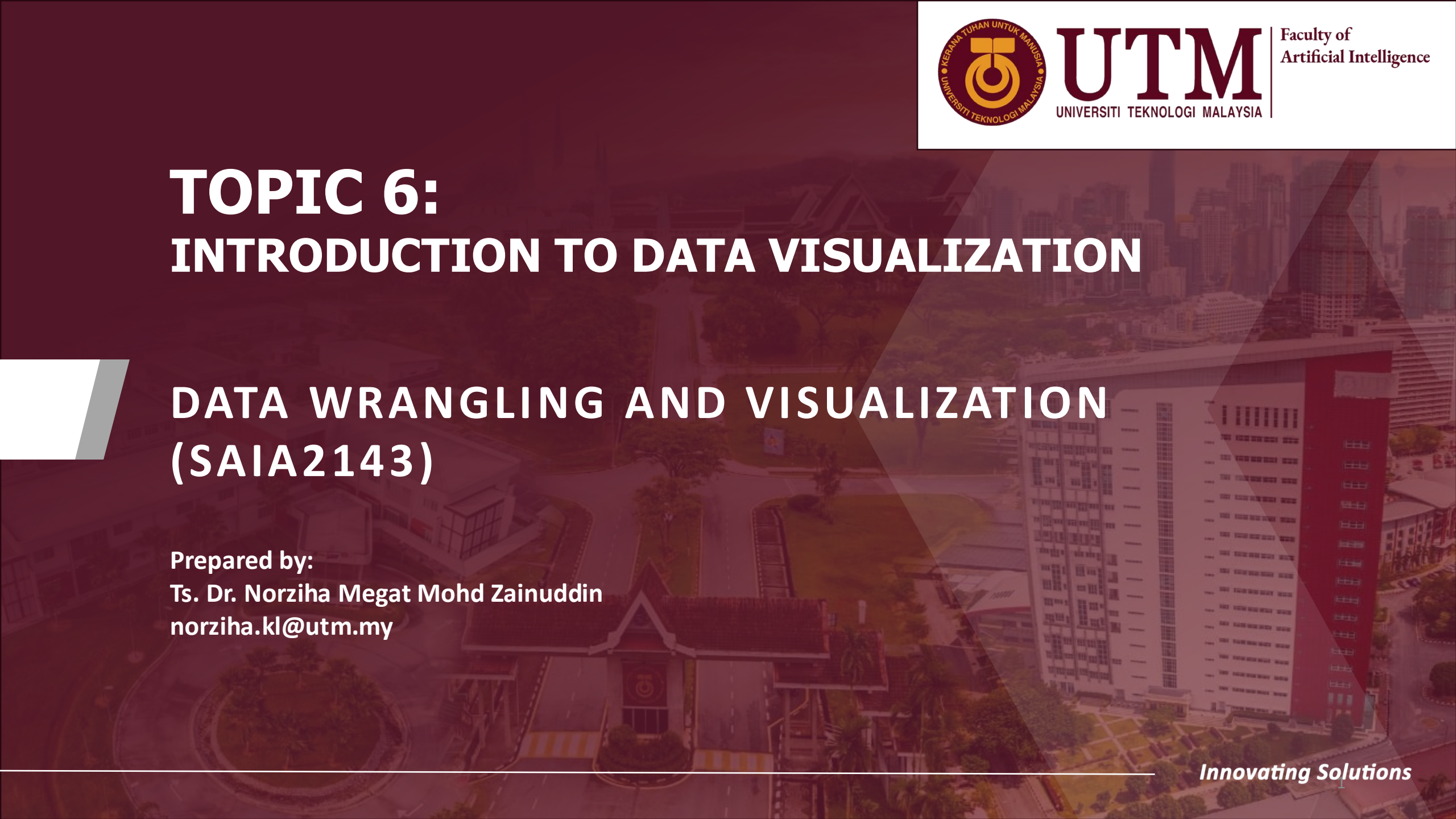


UTM
UNIVERSITI TEKNOLOGI MALAYSIA
Faculty of Artificial Intelligence

TOPIC 6:
INTRODUCTION TO DATA VISUALIZATION

DATA WRANGLING AND VISUALIZATION (SAIA2143)

Prepared by:
Ts. Dr. Norziha Megat Mohd Zainuddin
norziha.kl@utm.my

COURSE LEARNING OUTCOMES

At the end of this topic, students should be able to:

1. Explain why data visualization is essential in analysis.
2. Identify and describe common types of visualizations.
3. Apply principles of effective visualization to create clear and insightful charts.

THINK



What is visualization?

Topic 6: Introduction to Data Visualization

a. **Importance of visualization in data analysis**
b. Basic visualization concepts: charts, graphs, and plots
c. Principles of effective visualization

a. Why Visualization?

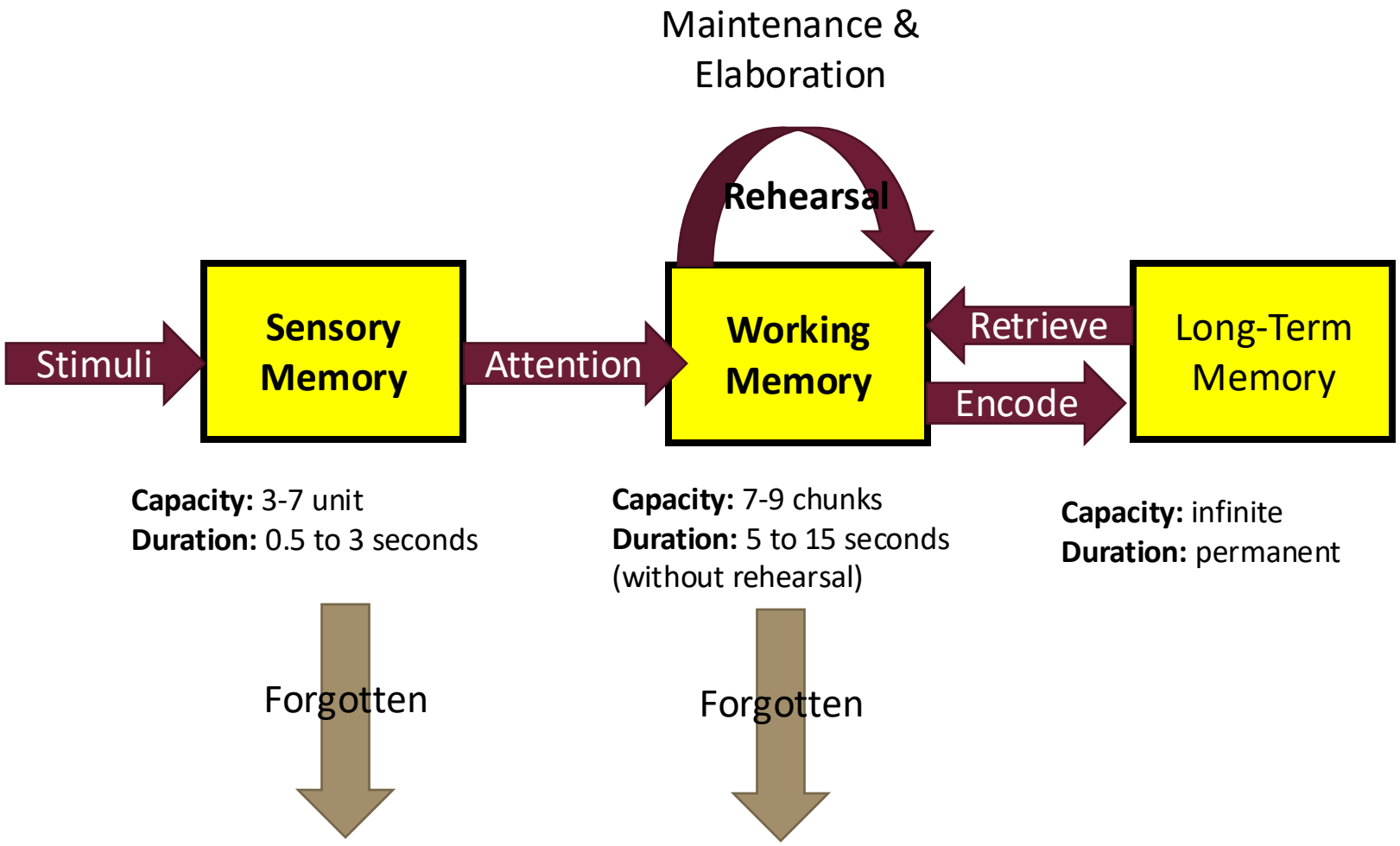
- Raw data is hard to interpret.
- Visualization helps summarize, simplify, and structure data.
- Humans process visuals 60,000x faster than text.
- Helps identify what's important at a glance.



Type of Memory

- Iconic memory or sensory memory
- Working memory or short term memory
- Long term memory

Type of memory



Type of Memory	Iconic memory or Sensory memory	Working memory	Long term memory
Duration	About a second	About a minutes	A second to a lifetime
How Processing Happens	Pre-attentive processing even before we pay attention	Can hold and process between 7-9 chunks of information	Information is store by repeated application or through rehearsal
Capacity	3-7 units	7-9 chunks	Infinite
Duration	0.5 to 3 seconds	5 to 15 seconds without rehearsal	Permanent

An exercise in pre-attentive processing

- How many 3s do you see?

1 0 4 2 1 3 7 8 6 7 6 4 3 2 1 4 6 7 5 4 6 8 8 6 5 9 3 0

1 4 6 7 5 4 6 8 8 6 5 9 3 0 7 5 6 5 7 6 7 8 7 8 3 7 8 7

4 3 6 7 5 7 8 7 6 7 7 8 9 8 1 0 4 2 1 3 7 8 6 7 6 4 3 2

7 5 6 5 7 6 7 8 7 8 3 7 8 7 6 8 8 6 5 9 3 0 7 5 6 5 7 6

An exercise in pre-attentive processing

- How many 3s do you see?

1 0 4 2 1 3 7 8 6 7 6 4 3 2 1 4 6 7 5 4 6 8 8 6 5 9 3 0

1 4 6 7 5 4 6 8 8 6 5 9 3 0 7 5 6 5 7 6 7 8 7 8 3 7 8 7

4 3 6 7 5 7 8 7 6 7 7 8 9 8 1 0 4 2 1 3 7 8 6 7 6 4 3 2

7 5 6 5 7 6 7 8 7 8 3 7 8 7 6 8 8 6 5 9 3 0 7 5 6 5 7 6

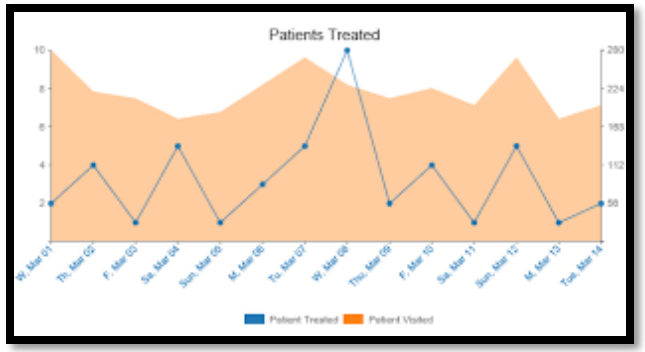


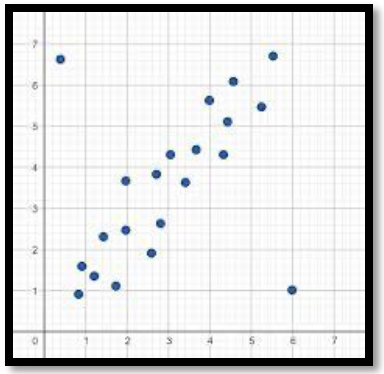
Pre-attentive Processing

- Don’t really have time to think - ten 3s.
- Intensity of colour to make our brain quick to pick up.
- < 200–250 ms qualifies as pre-attentive
  - eye movements take at least 200ms
  - yet certain processing can be done very quickly, implying low-level processing in parallel
- Pre-attentive attributes help us ***enable our audience to see what we want them to see before they even know they’re seeing it!***

b. Patterns, Trends, and Anomalies at a Glance

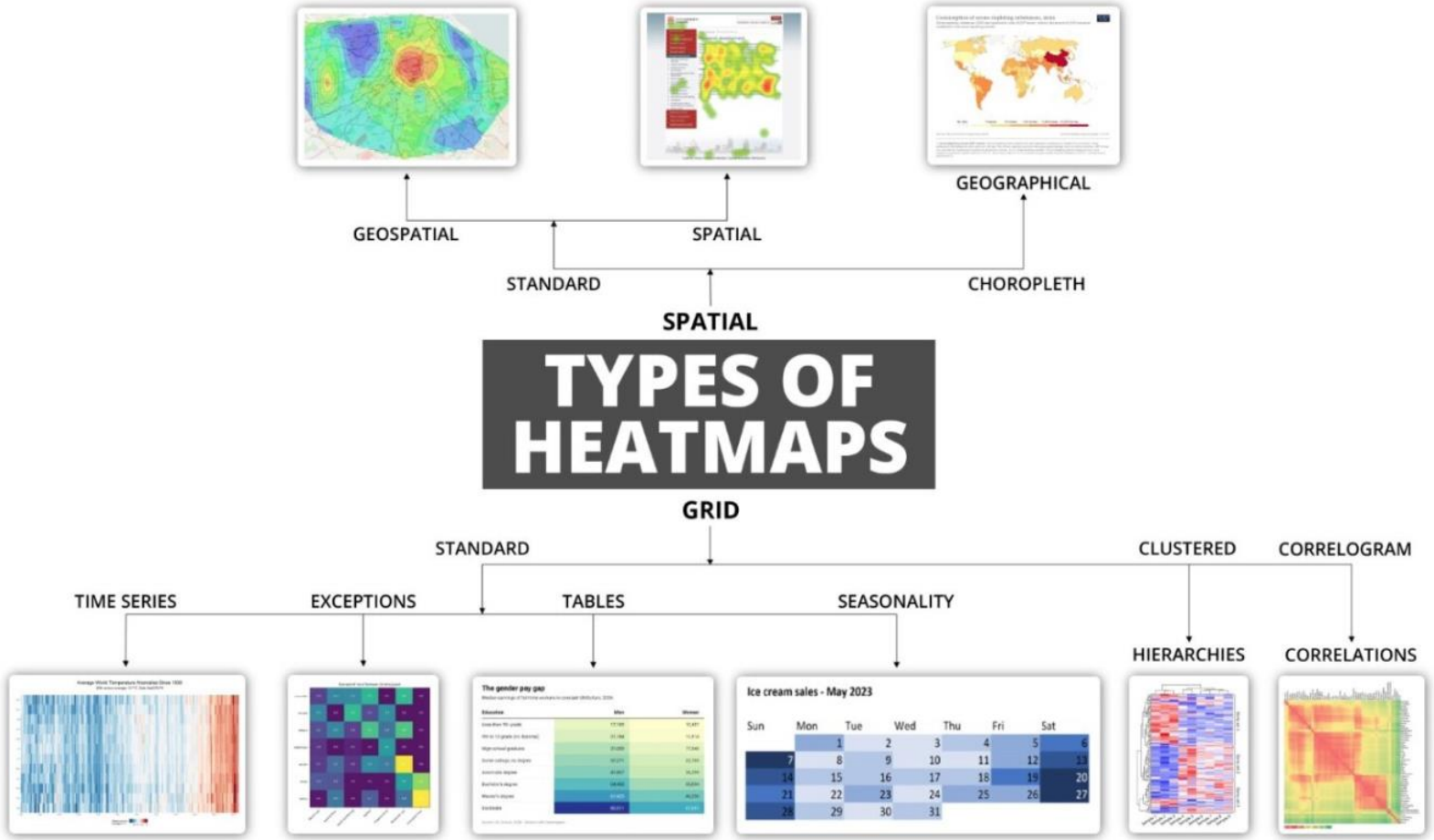
- Line charts show trends over time.
- Scatter plots show relationships and clusters.
- Outliers become obvious in charts but are hidden in tables.
- Heatmaps show intensities and distributions.





	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Average Monthly Temperatures at Central Park, New York													
2		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
3	2016	34.5	37.7	48.9	53.3	62.8	72.3	78.7	79.2	71.8	58.8	49.8	38.3	
4	2017	38.0	41.6	39.2	57.2	61.1	72.0	76.8	74.0	70.5	64.1	46.6	33.4	
5	2018	31.7	42.0	40.1	49.5	66.9	71.7	77.6	78.1	70.7	57.7	44.4	40.1	
6	2019	32.5	36.2	41.7	55.5	62.2	71.7	79.6	75.5	70.4	59.9	43.9	38.3	
7	2020	39.1	40.1	48.0	50.4	60.3	73.7	80.0	76.9	68.8	57.9	53.0	39.2	
8	2021	34.8	34.2	45.8	54.6	62.9	74.3	76.0	77.5	70.3	62.0	46.2	43.8	
9	2022	30.3	37.3	45.3	52.8	64.0	71.4	79.5	79.3	69.5	56.9	50.9	38.5	
10	2023	43.5	41.1	44.6	57.6	62.7	70.0	79.0	75.0	69.4	60.5	46.7	44.6	
11	2024	37.0	40.1	48.1	55.6	65.0	75.1	79.6	75.1	68.8	60.9	51.4	38.2	
12														

Heatmaps in Data Visualization

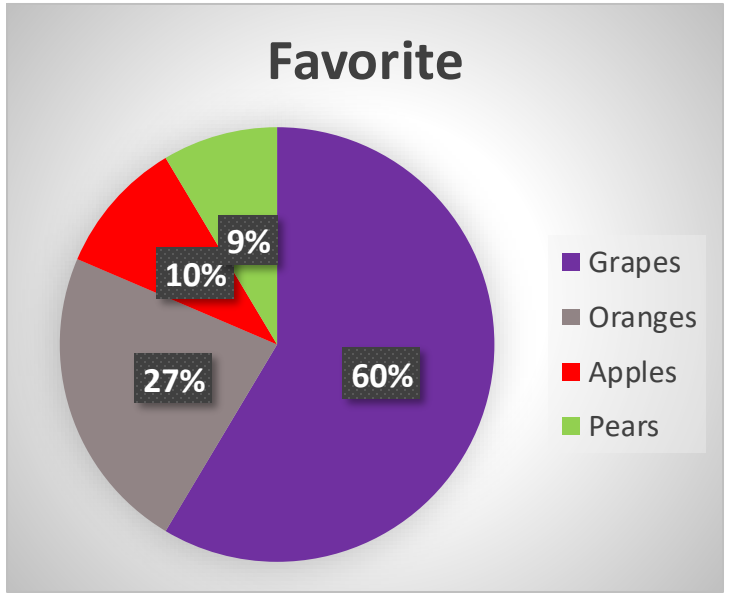


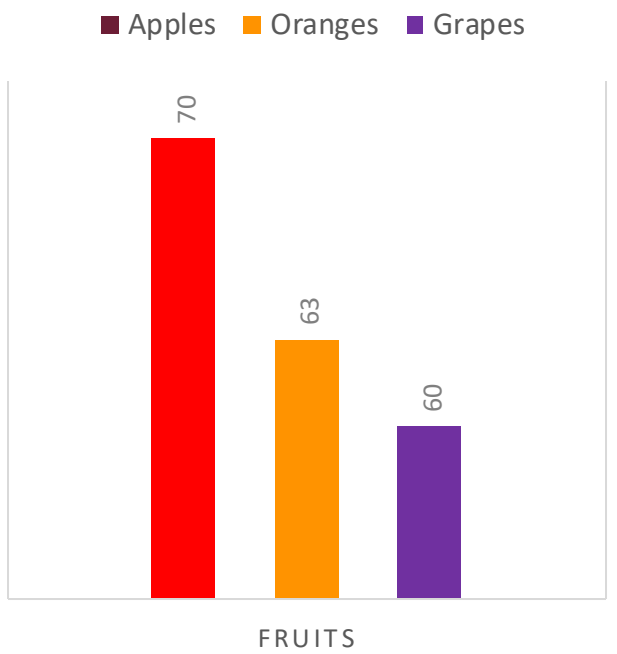
c. Enhance Communication and Decision-Making

- Decision-makers need clarity, not complexity.
- Visualization helps non-technical audiences grasp data-driven conclusions.
- Used in meetings, dashboards, reports.
- Supports persuasive storytelling with evidence.



d. Good vs. Bad Visualizations





- Good visualizations are clear, focused, and honest.
- Bad visualizations can mislead with scale, colors, 3D effects.
- Key message: effectiveness over flashiness.

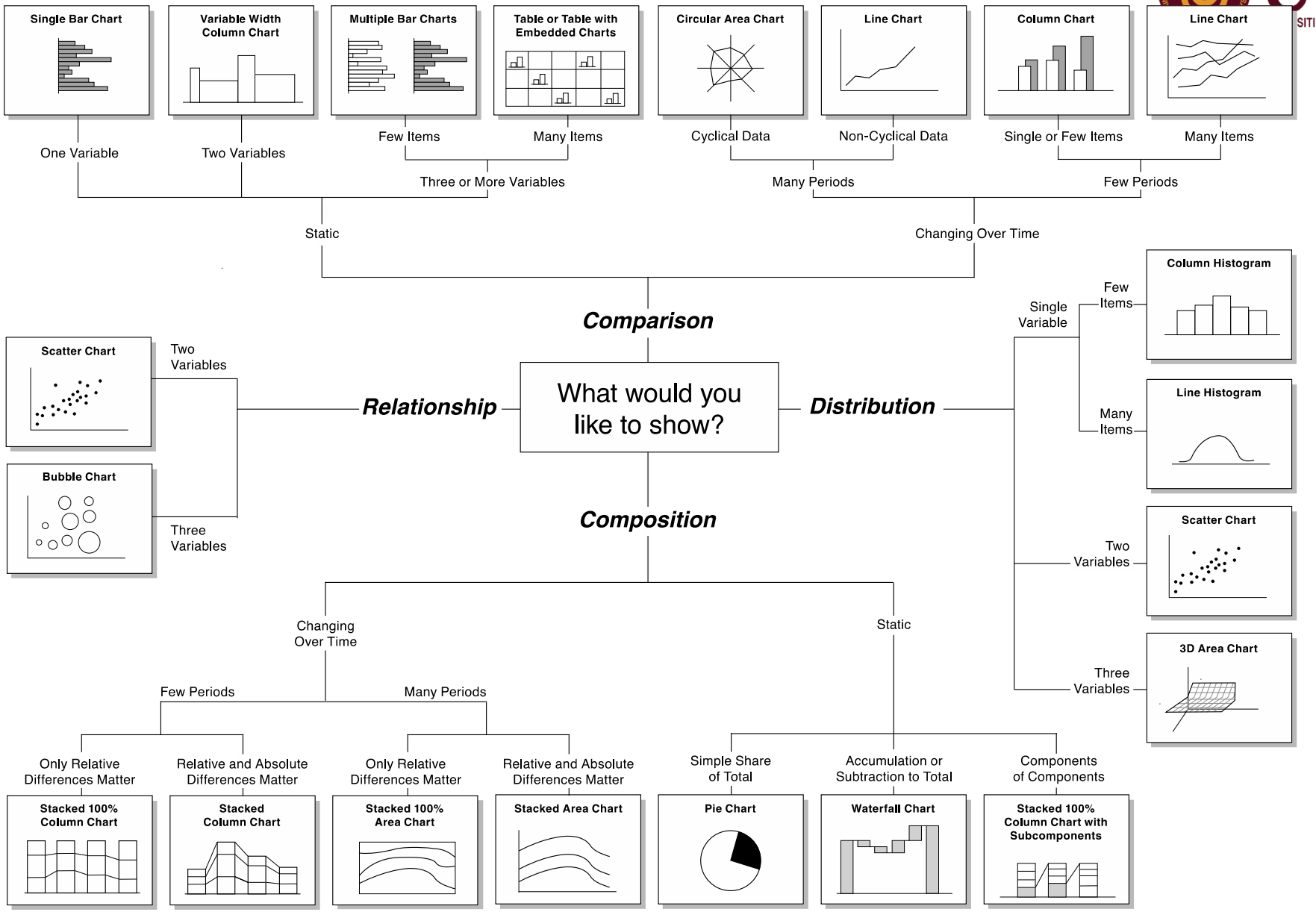
Topic 6: Introduction to Data Visualization

a. Importance of visualization in data analysis
b. **Basic visualization concepts: charts, graphs, and plots**
c. Principles of effective visualization

Choosing the Right Visualization

Question	When to Use?	Suggested Chart	Example
Comparing categories?	Comparing sales by region, product quantities, or survey results.	Bar Chart	Number of students enrolled in different majors.
Showing a trend over time?	Stock prices over months, temperature changes, monthly sales (Often used for time-series data)	Line Chart	Website traffic over 12 months.
Showing parts of a whole?	Simple data with few categories.	Pie Chart	Market share of different smartphone brands.
Understanding distribution?	Analyzing exam scores, income brackets, or age groups. Visualize the distribution, median, quartiles, and outliers.	Histogram / Box Plot	Distribution of student test scores.
Exploring relationships?	Identifying correlation, trends, or outliers.	Scatter Plot	Height vs. weight of students.

Chart Chooser

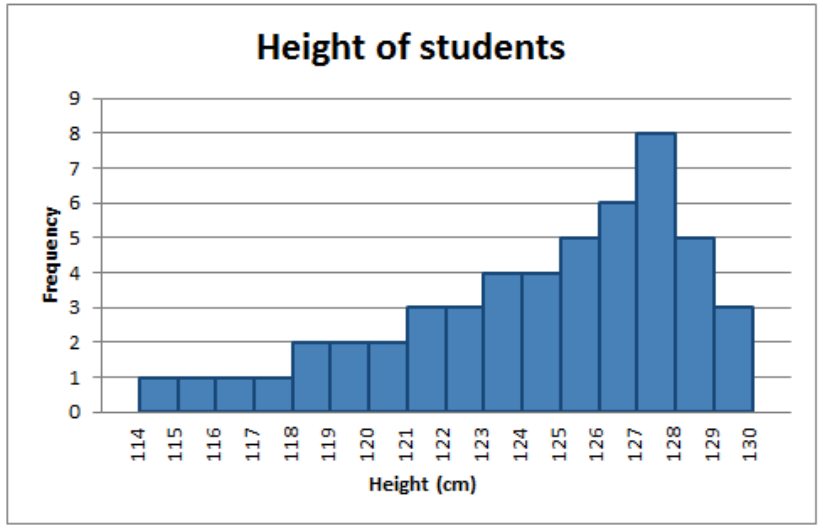


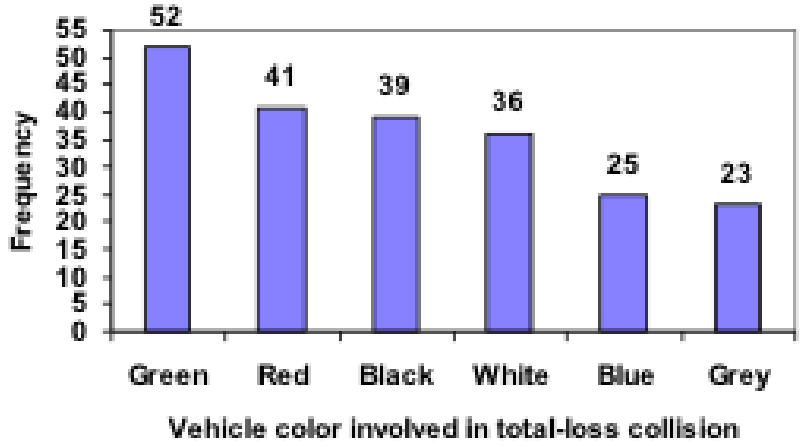
© 2020 Andrew V. Abela, Dr.Abela@ExtremePresentation.com
www.extremepresentation.com

www.fai.utm.my

Pre-attentive attributes in graphs

- Quantitative data - graph to represent the values.
- Pre-attentive visual attributes in both quantitative and categorical.

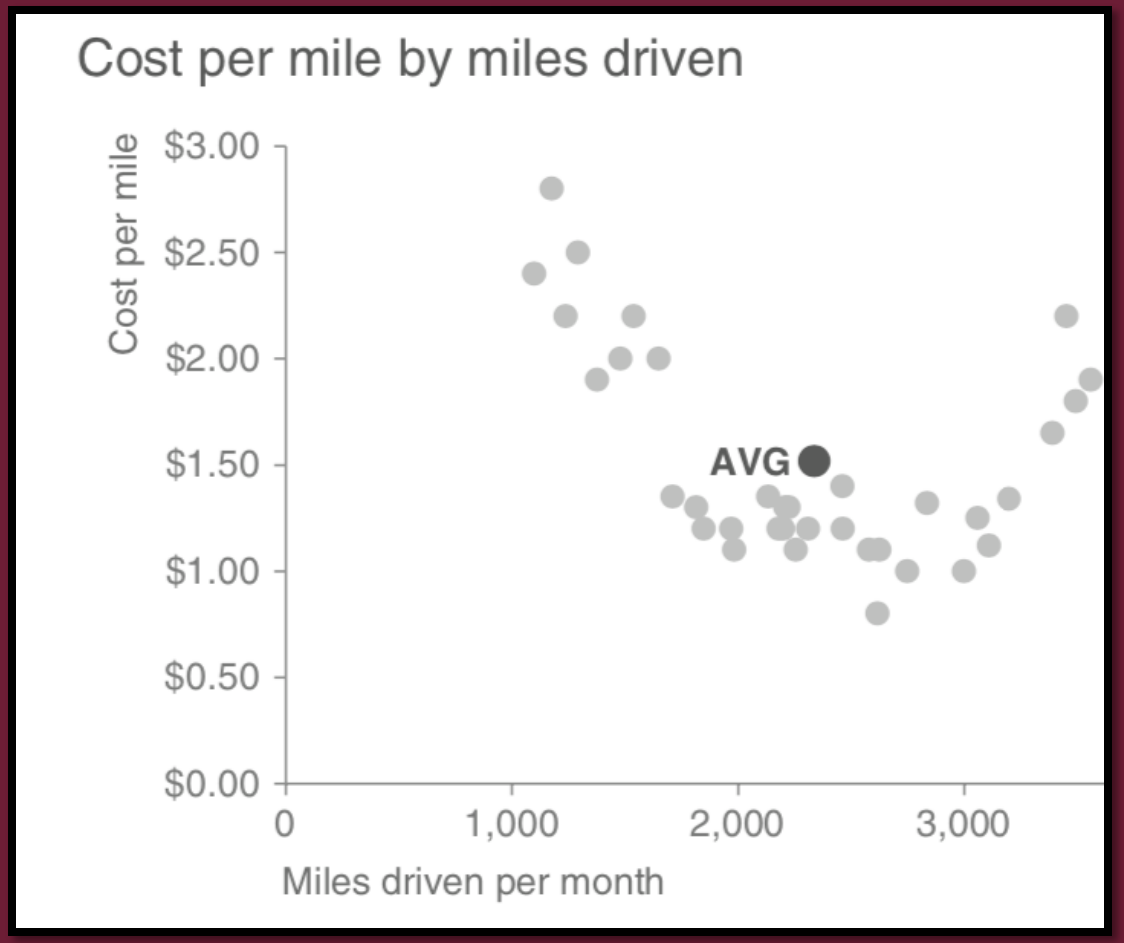




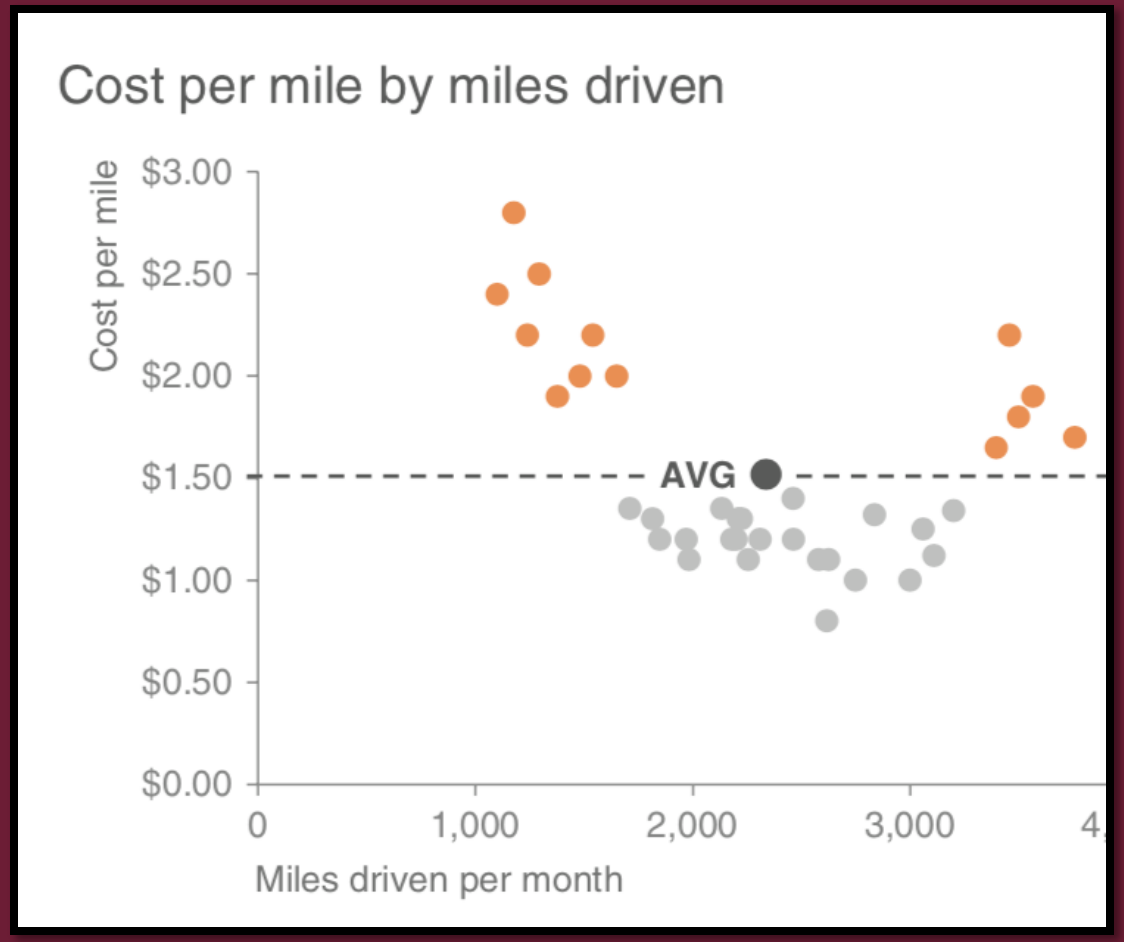
Pre-attentive attributes in graphs...

a. Points:

Small circles or polygons that represent values based on their 2-D position in relation to a quantitative scale along one or both of the axes



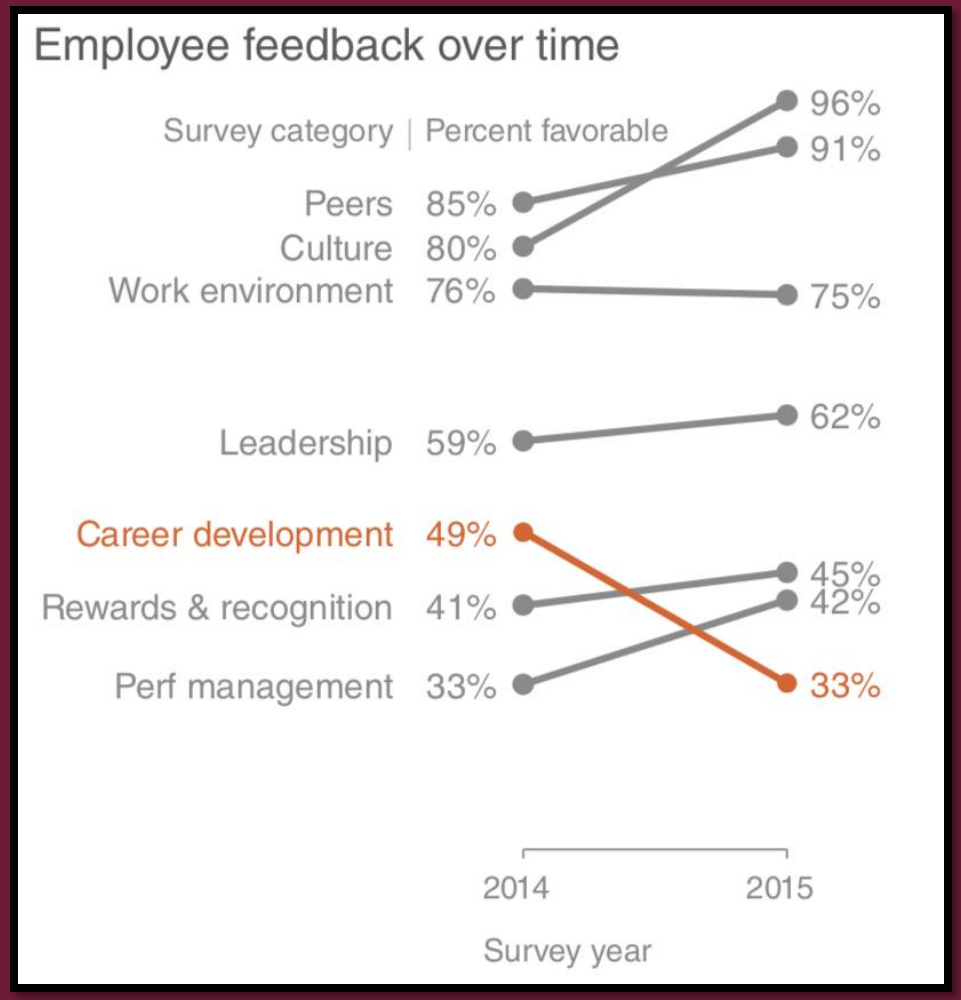
Pre-attentive attributes in graphs...

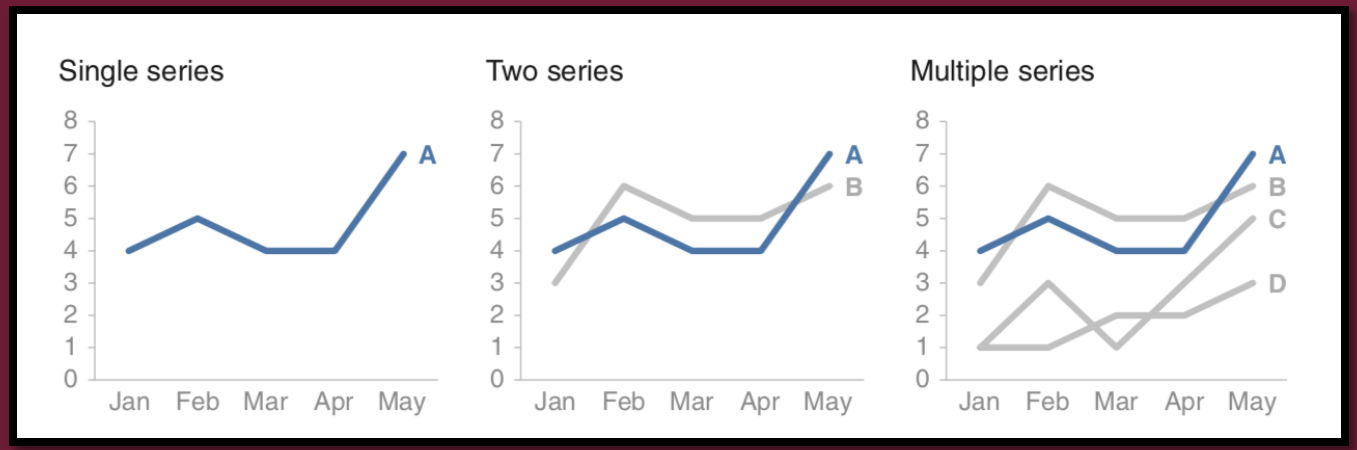


b. Lines:

Typically used to connect a series of 2-D positions to represent a series of values, relative to a quantitative scale along one of the axes

Pre-attentive attributes in graphs...

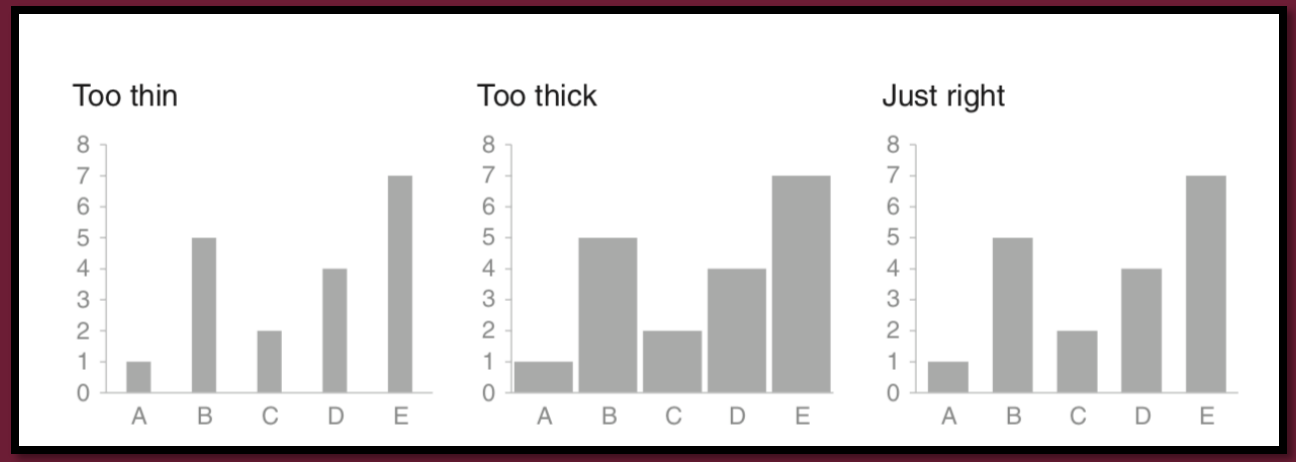


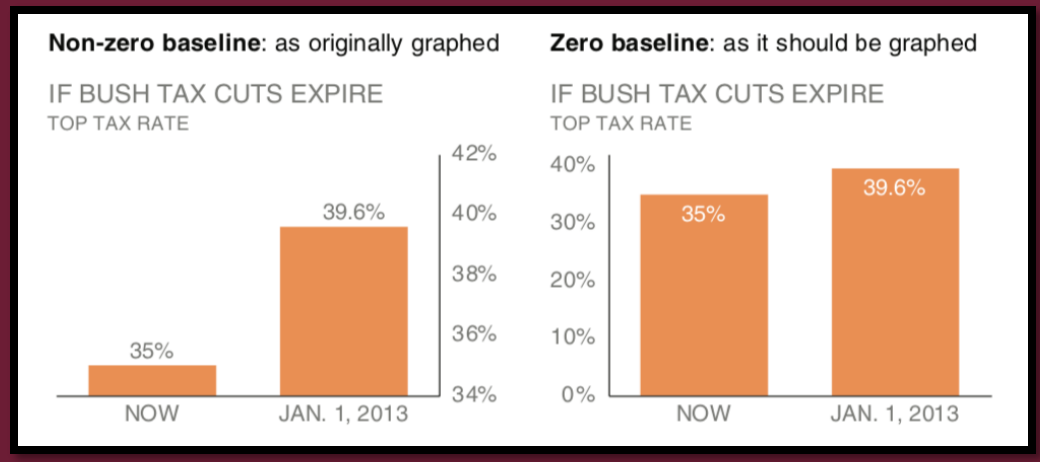


Pre-attentive attributes in graphs...

c. Rectangles:

Usually called bars, and for specific purposes boxes, these vary in length only and represent values based both on their lengths and the 2-D positions of their ends in relation to a quantitative scale along one of the axes

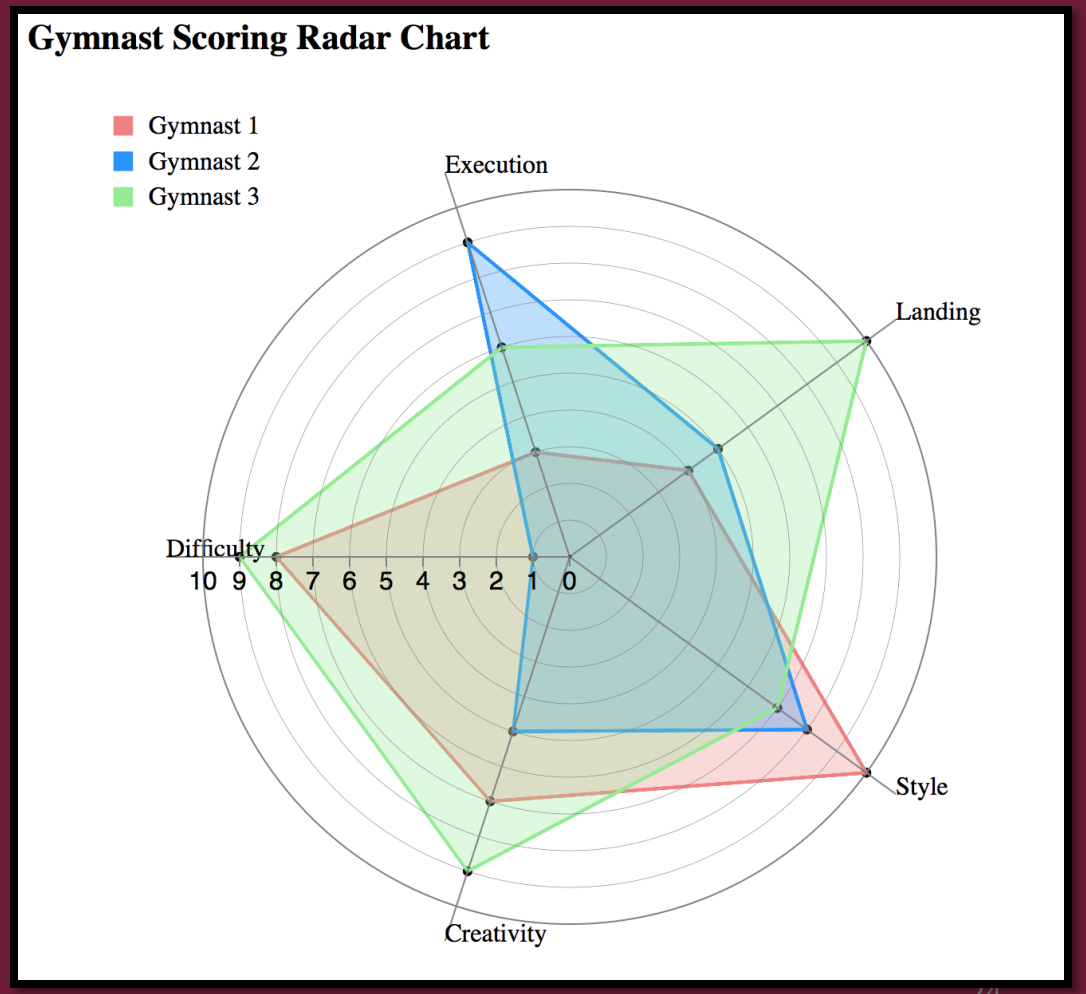




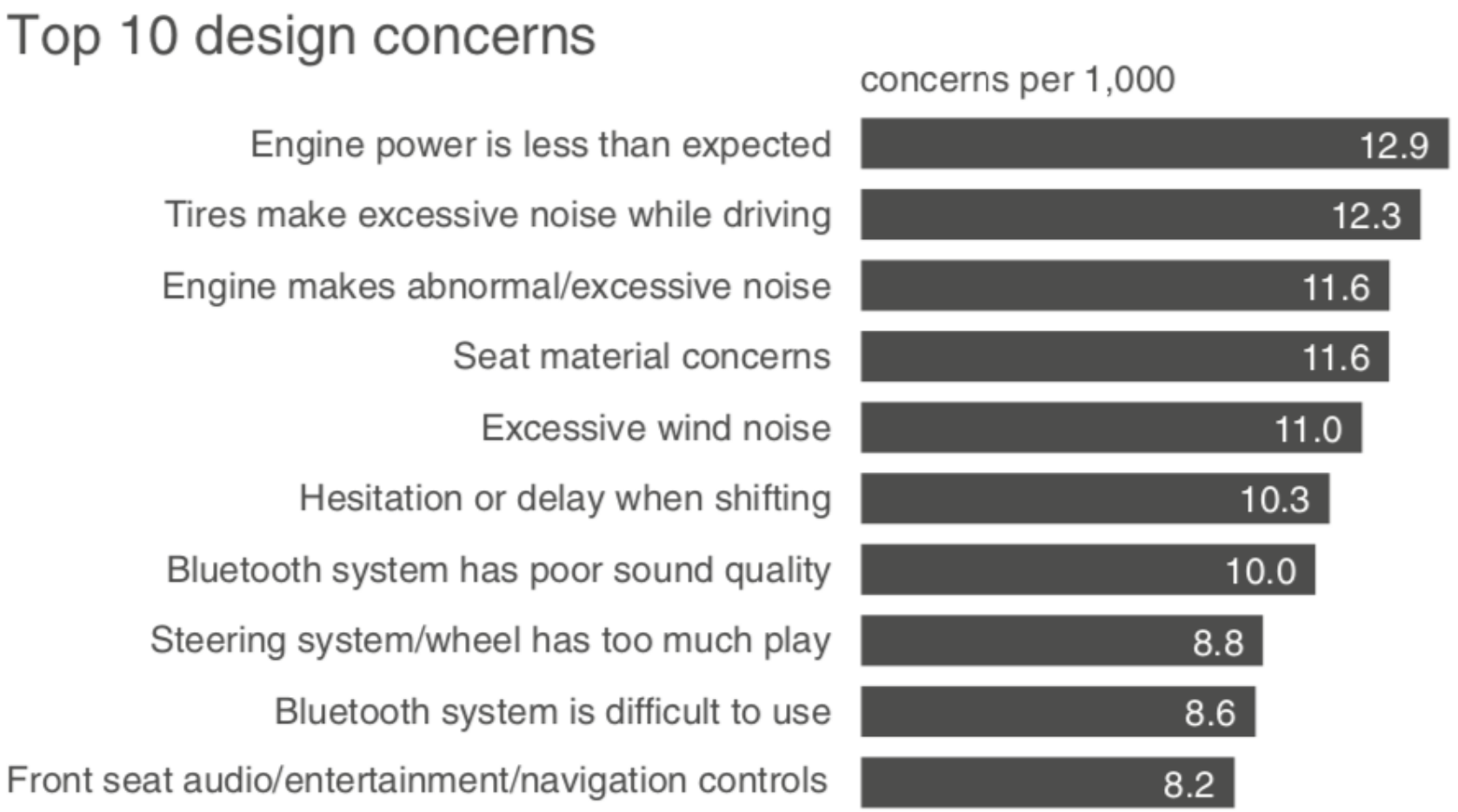
Pre-attentive attributes in graphs...

d. Areas:

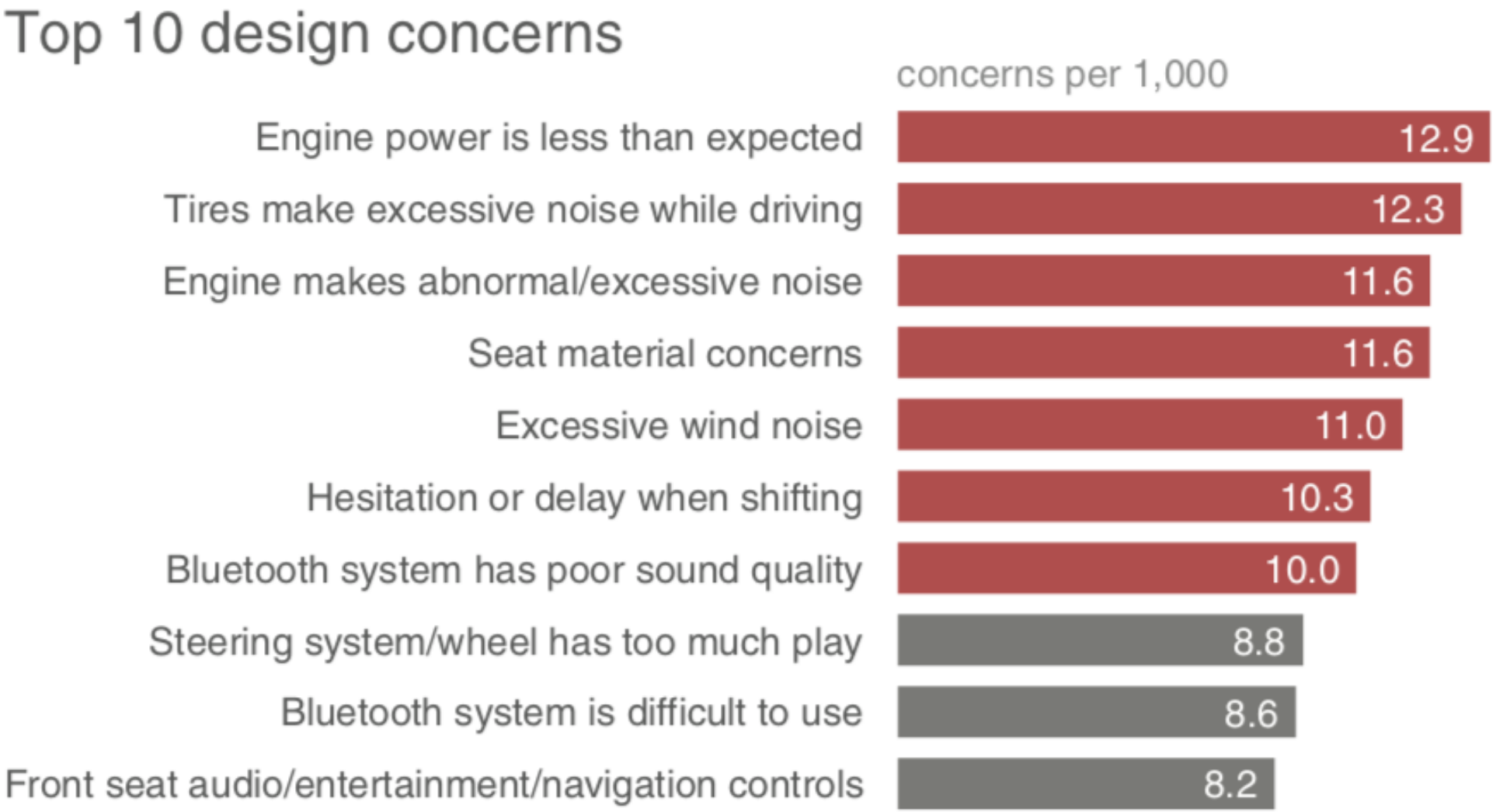
Circles or polygons that represent values based on their areas



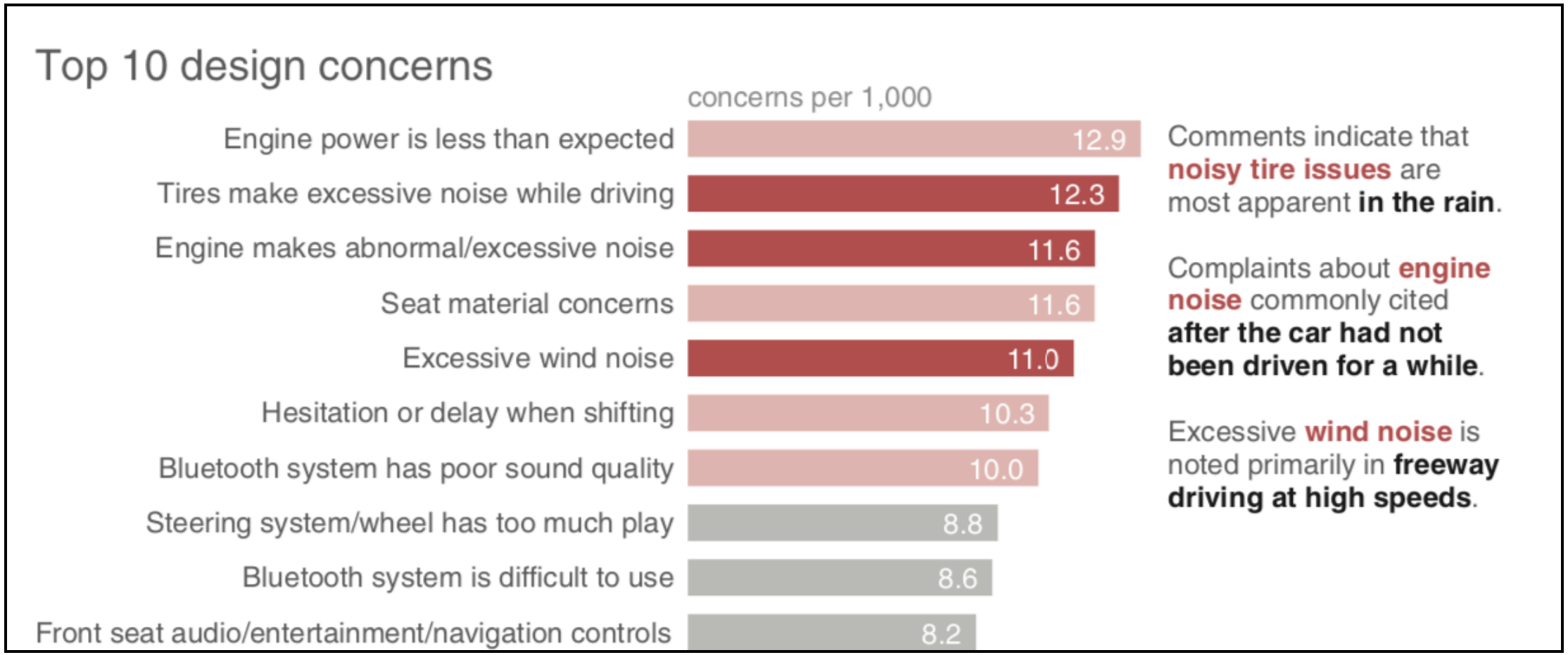
Original graph no Pre-attentive attributes



Leverage color to draw attention



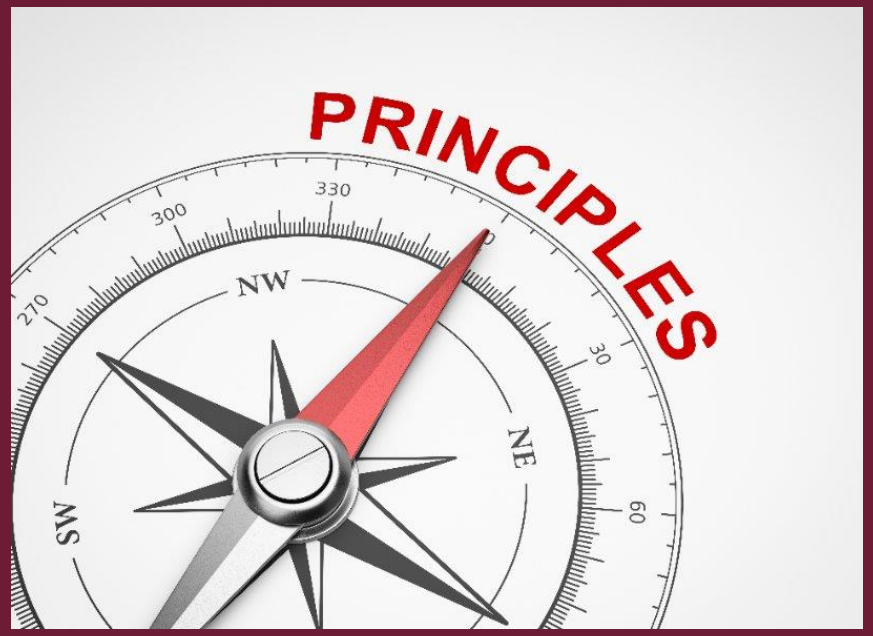
Create a visual hierarchy of information



Topic 6: Introduction to Data Visualization

a. Importance of visualization in data analysis
b. Basic visualization concepts: charts, graphs, and plots
c. **Principles of effective visualization**

i. Why Do We Need Principles?



Good visuals tell a story.

Poor visuals confuse or mislead.

Principles help us:

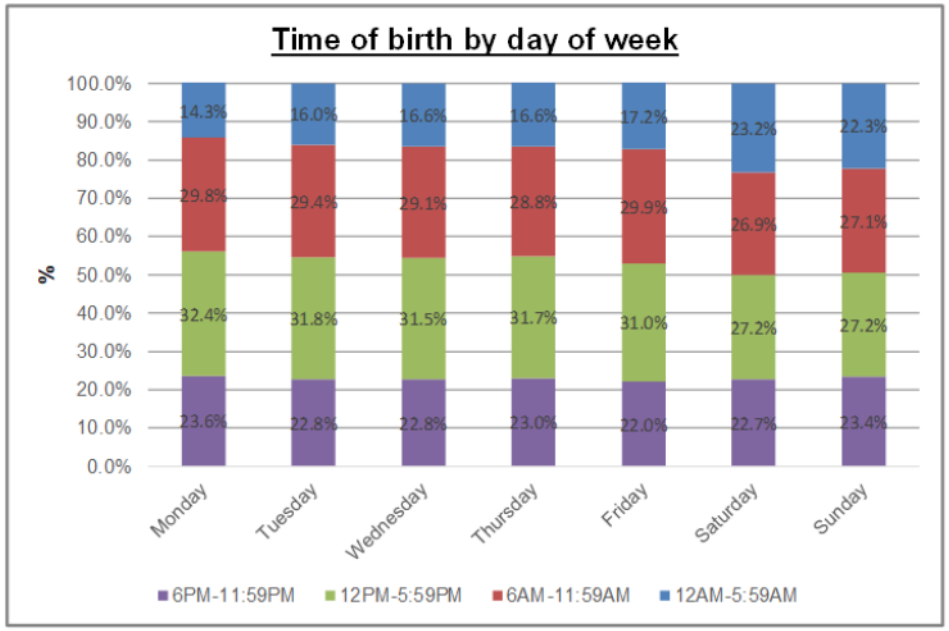
- Enhance clarity
- Avoid distortion
- Maintain viewer trust

ii. Clarity Over Complexity

- Focus on the **main message**.
- Use **simple, readable fonts**.
- Avoid clutter: gridlines, heavy borders, **too many colors**.

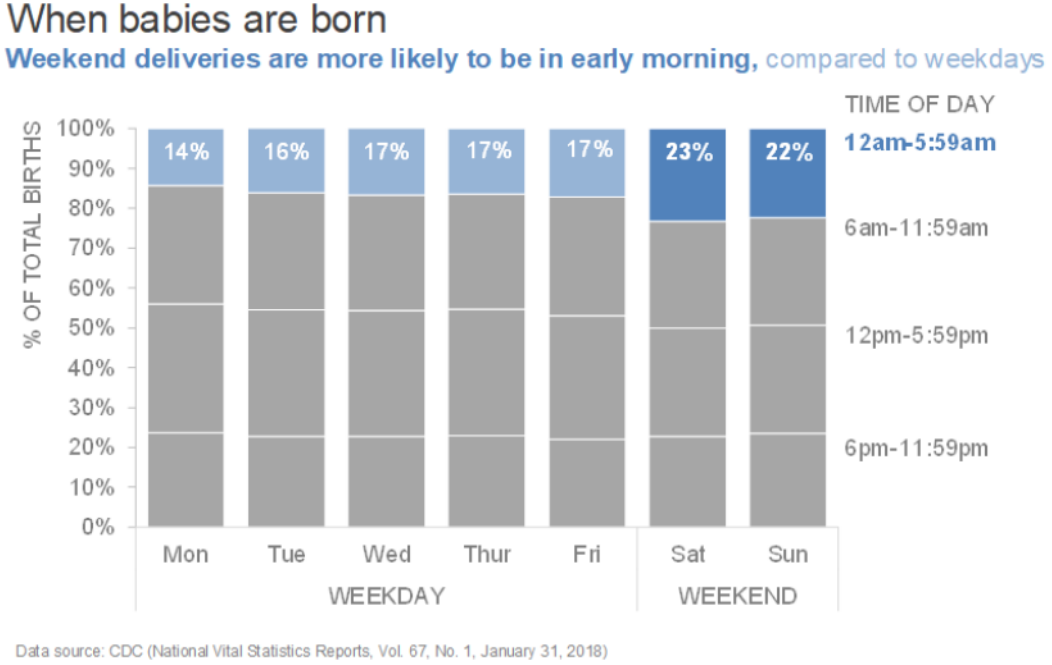
Example

BEFORE



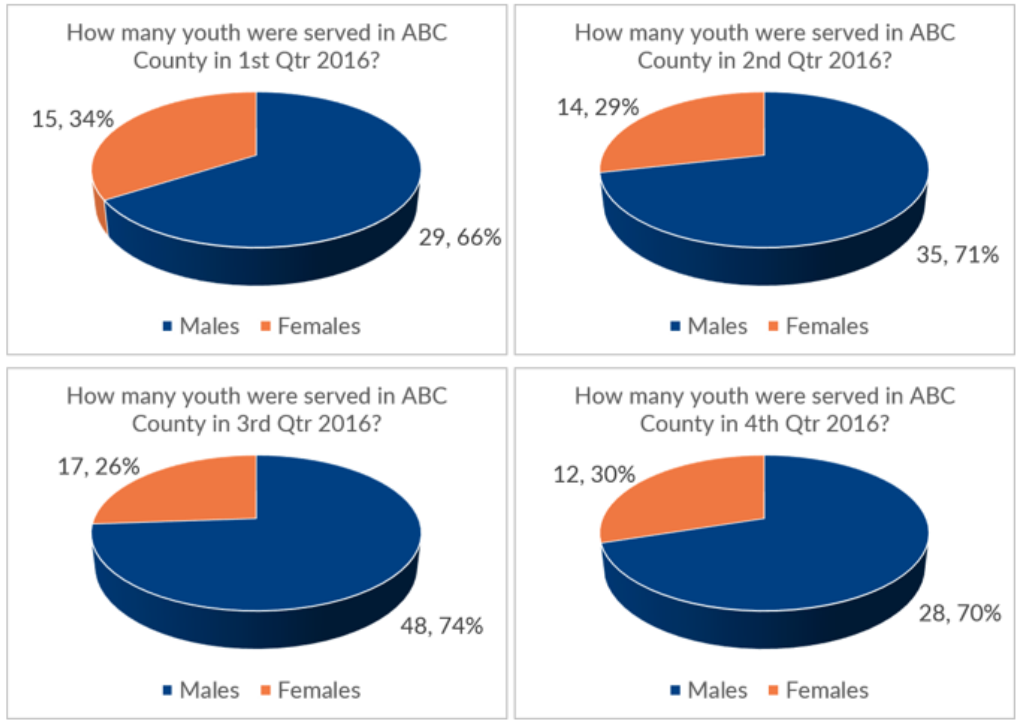
Data source: CDC (National Vital Statistics Reports, Vol. 67, No. 1, January 31, 2018)

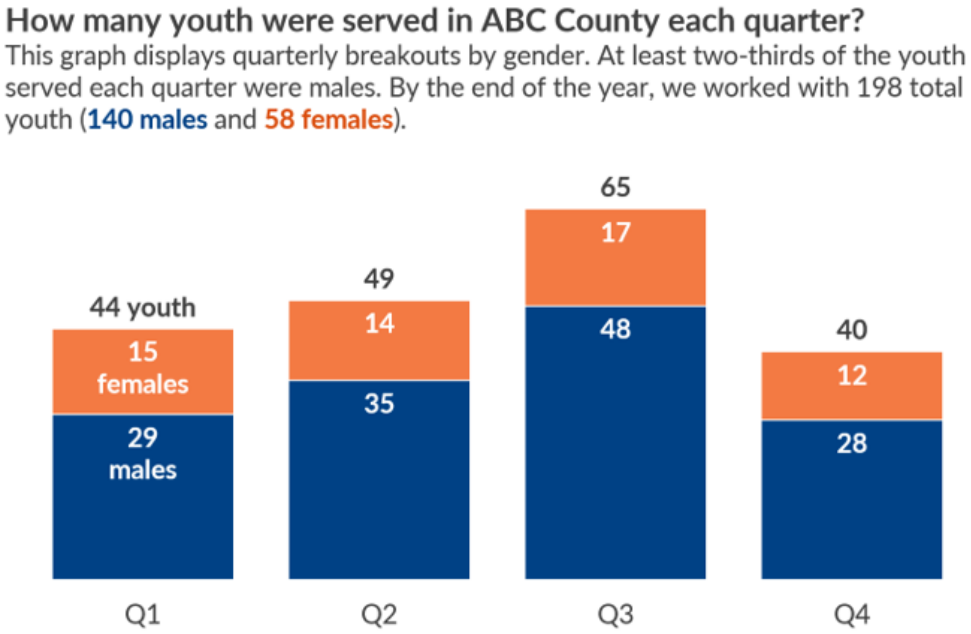
AFTER



Data source: CDC (National Vital Statistics Reports, Vol. 67, No. 1, January 31, 2018)

Example


How many youth were served in ABC County in 1st Qtr 2016?
15, 34%
29, 66%
Males
Females
How many youth were served in ABC County in 2nd Qtr 2016?
14, 29%
35, 71%
Males
Females
How many youth were served in ABC County in 3rd Qtr 2016?
17, 26%
48, 74%
Males
Females
How many youth were served in ABC County in 4th Qtr 2016?
12, 30%
28, 70%
Males
Females


How many youth were served in ABC County each quarter?
This graph displays quarterly breakouts by gender. At least two-thirds of the youth served each quarter were males. By the end of the year, we worked with 198 total youth (140 males and 58 females).
44 youth
15 females
29 males
49
14
35
65
17
48
40
12
28
Q1
Q2
Q3
Q4

iii. Use Color with Purpose

- Use **contrast** to highlight key data.
- Avoid using **too many colors**—stick to a palette.
- Be mindful of **colorblind accessibility**.

Importance of Color in Data Visualization

a. Highlights patterns and insights
b. Enhances readability
c. Differentiates groups or categories
d. Supports accessibility and communication

Properties of Color

Hue determines the base color.

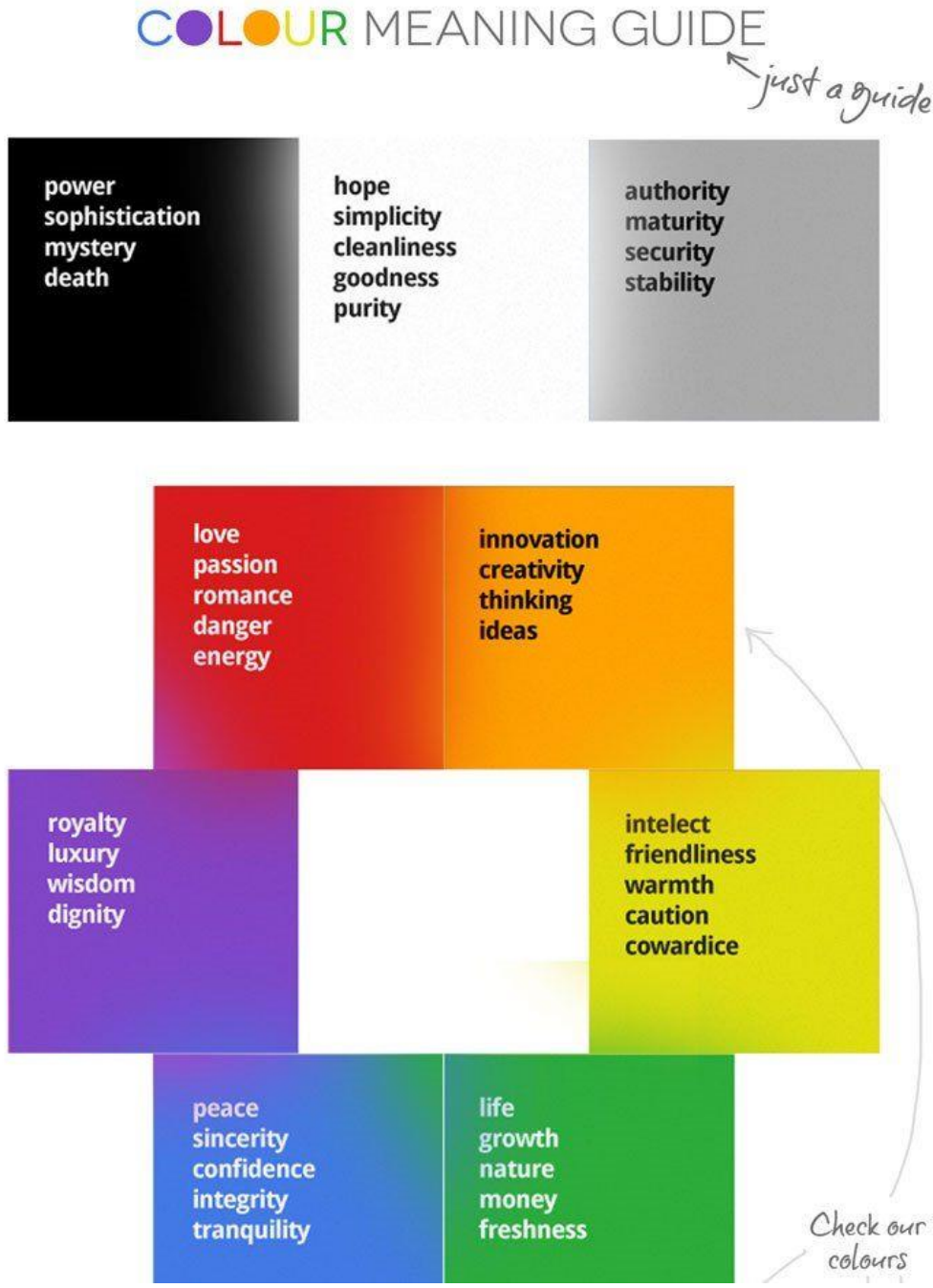
Saturation defines how pure or diluted that base color is.

Luminance controls how bright or dark the color appears.

Colour Meaning Guide

Website link:

https://www.iconvisual.com.au/creative/the-meaning-of-colour-in-creative-design

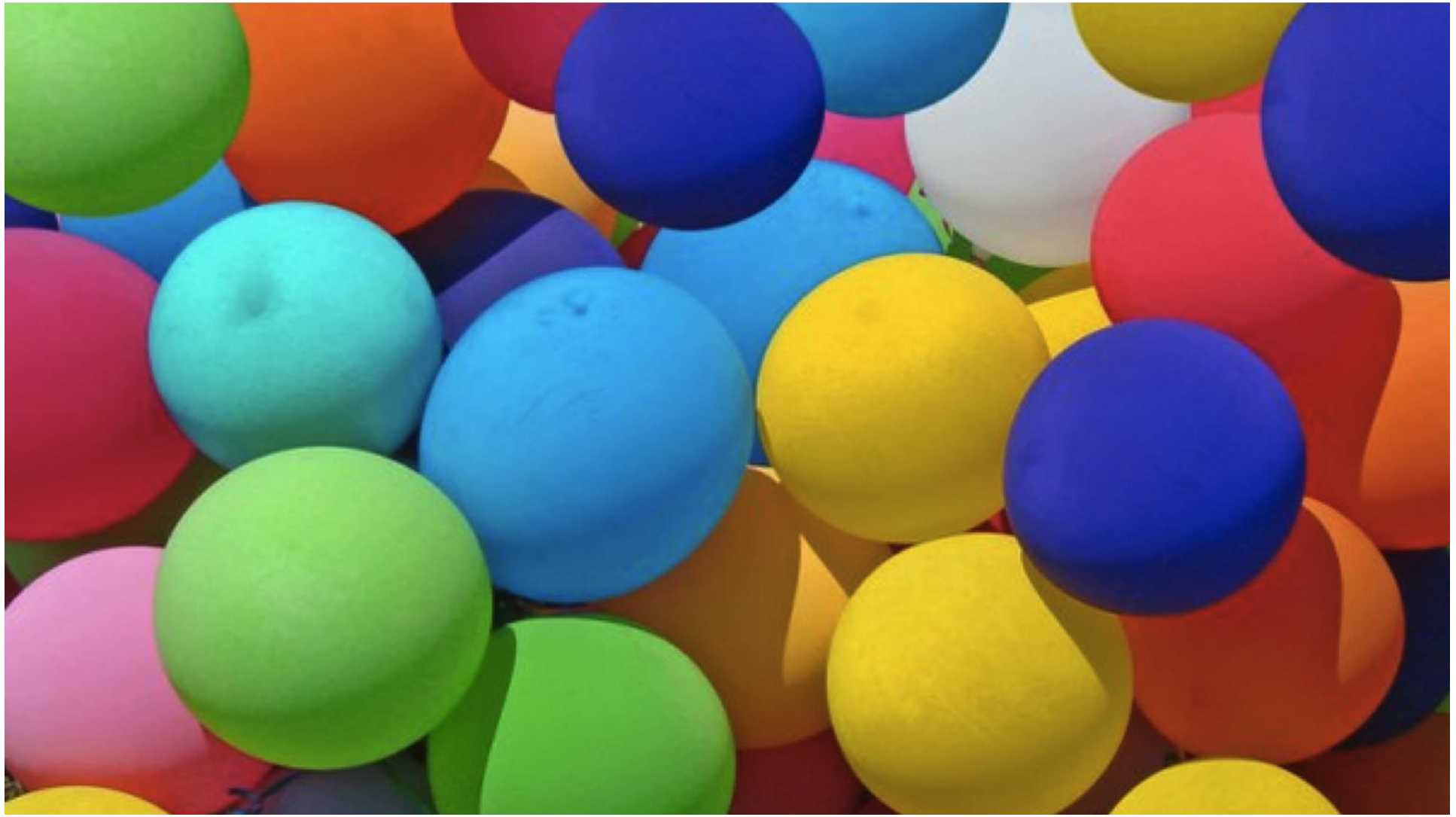




Where are your eyes drawn?




12
60
FAMILY DOLLAR
Garage Sale
329 Walnut St
GARAGE SALE
SAUK PRAIRIE SOCCER CAMP
608-963-9422
GARAGE SALE
Perennial Sale
GARAGE SALE
GARAGE SALE
1015 Van Buren St.
SALE!
CLOTHES, TOOLS, TOYS, ELECTRONICS
KIDS STUFF, & MORE!!!
FRI-TUE 8am-6pm
683 9th Street
GARAGE SALE
THU 7AM FRI 9AM
175 Chestnut St.
157 · 5TH
SALE · TODAY
MOVING SALE
Merriage
(Across from Varsity)
GARAGE SALE





Color sparingly

Country Level Sales Rank Top 5 Drugs

Rainbow distribution in color indicates sales rank in given country from #1 (red) to #10 or higher (dark purple)

Country	A	B	C	D	E
AUS	1	2	3	6	7
BRA	1	3	4	5	6
CAN	2	3	6	12	8
CHI	1	2	8	4	7
FRA	3	2	4	8	10
GER	3	1	6	5	4
IND	4	1	8	10	5
ITA	2	4	10	9	8
MEX	1	5	4	6	3
RUS	4	3	7	9	12
SPA	2	3	4	5	11
TUR	7	2	3	4	8
UK	1	2	3	6	7
US	1	2	4	3	5

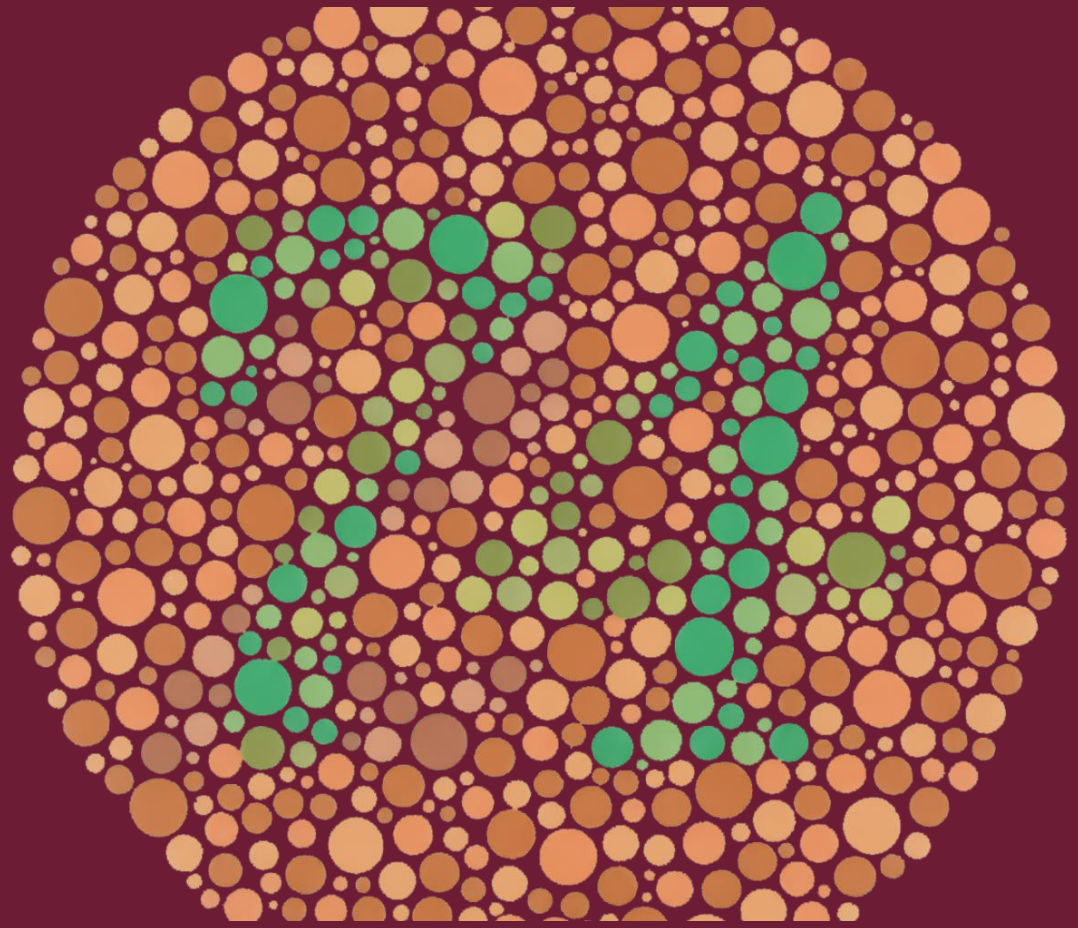
Top 5 drugs: country-level sales rank

RANK 1 2 3 4 5+

COUNTRY | DRUG

	A	B	C	D	E
Australia	1	2	3	6	7
Brazil	1	3	4	5	6
Canada	2	3	6	12	8
China	1	2	8	4	7
France	3	2	4	8	10
Germany	3	1	6	5	4
India	4	1	8	10	5
Italy	2	4	10	9	8
Mexico	1	5	4	6	3
Russia	4	3	7	9	12
Spain	2	3	4	5	11
Turkey	7	2	3	4	8
United Kingdom	1	2	3	6	7
United States	1	2	4	3	5

Color Blind



Avoid to use red and green in same time in your design

How COLORBLIND people see things!



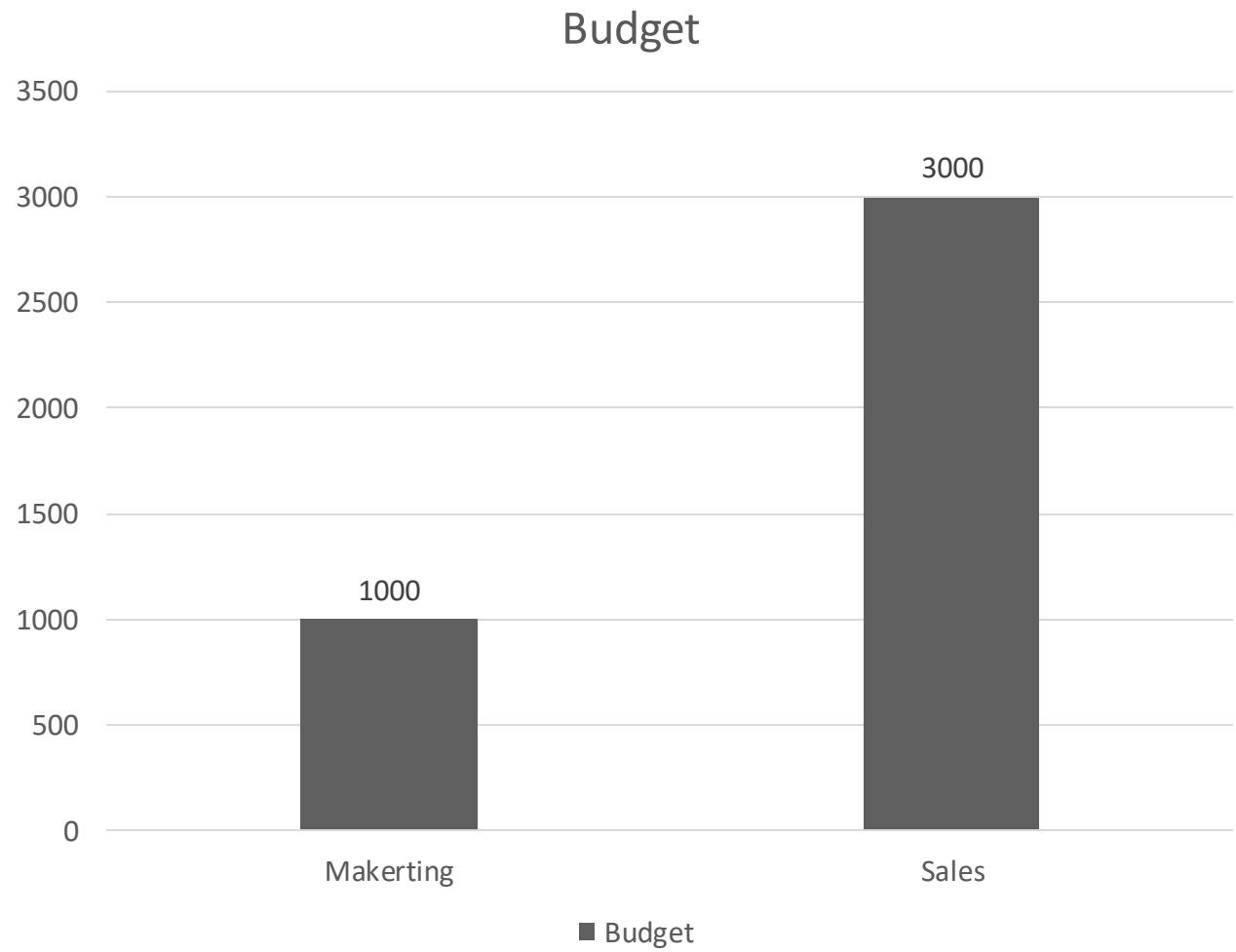


iv. Be Honest with Your Data

- Start axes at zero (unless otherwise noted).
- Don’t exaggerate differences with scale manipulation.
- Avoid 3D distortions or misleading proportions.

Visualizing Area

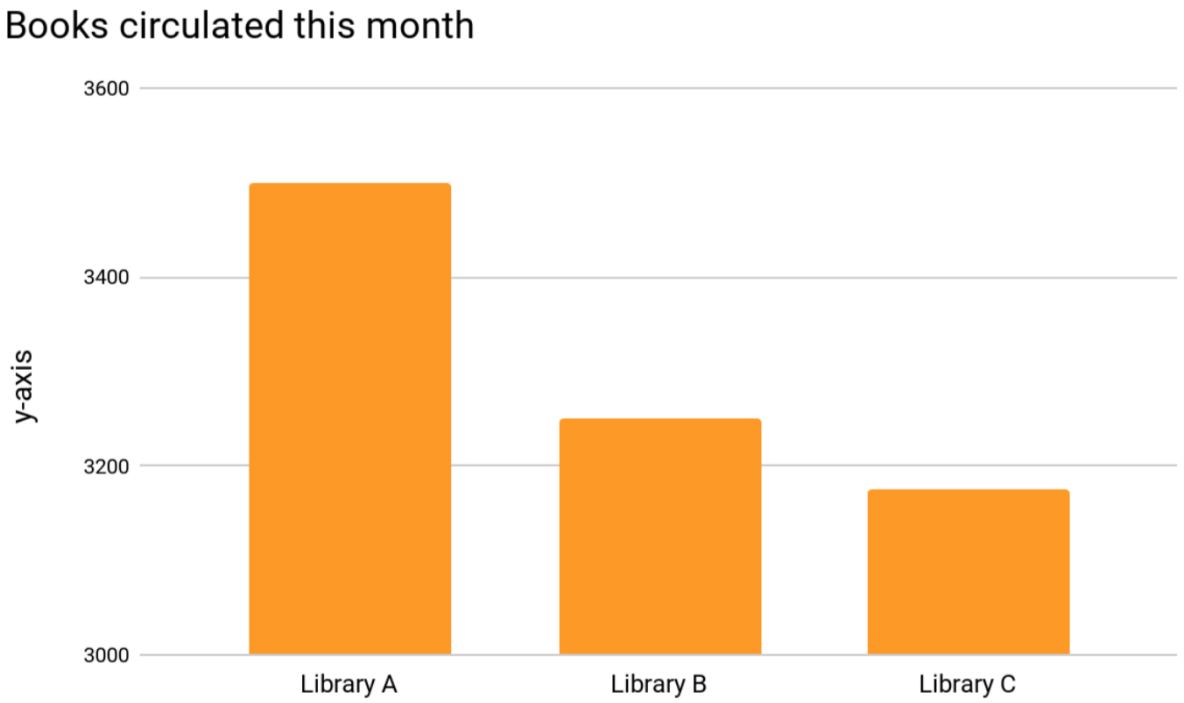
Each rectangles remains constant, only height changes



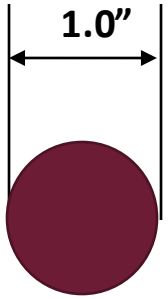
CLASS ATIVITY

Instruction:

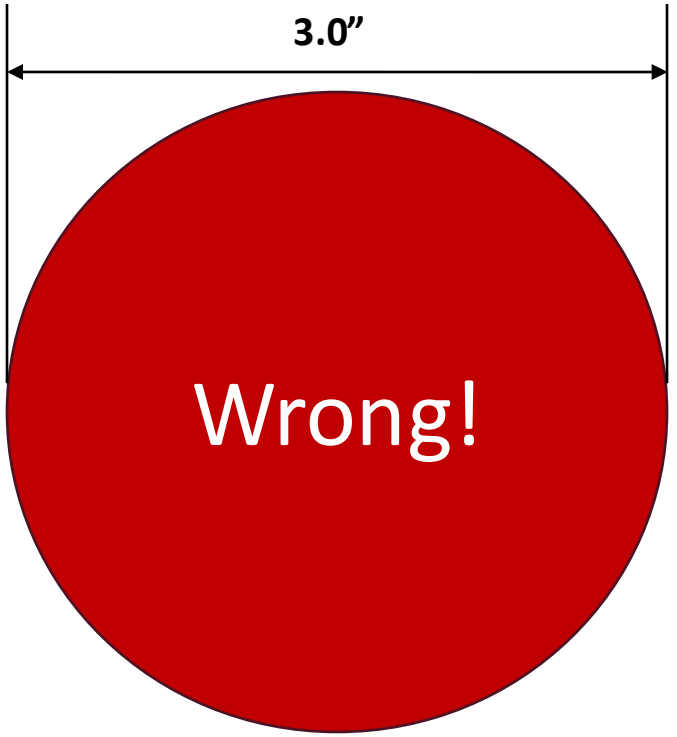
What is the mistake?



Visualizing Area…



$1M
Marketing
Budget
Area = 0.79in^2



$3M
Sales Budget
Area = 7.07in^2

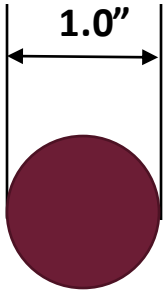
Circle Calculations Spreadsheet

Department	Value	Master Diameter	Master Radius	Area	Radius	Diameter
Marketing	$1M	1.0	0.5	0.7854		
Sales	$3M			0.3562	0.866	1.732

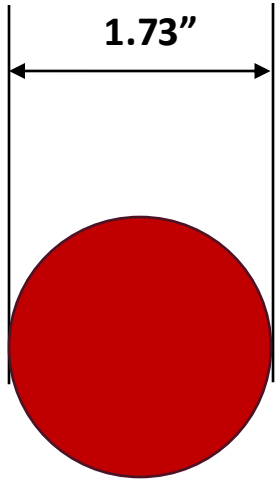
The equation to calculate the radius of a circle form the known area follows:

$$\text{Radius} = \sqrt{(\text{Area}/\pi)}$$

Visualizing Area…



$1M
Marketing Budget
Area = 0.79 in^2



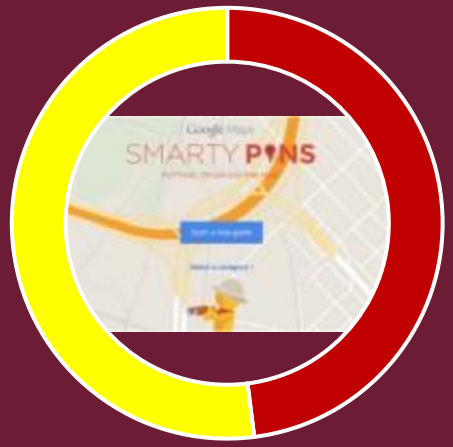
$3M
Sales Budget
Area = 2.36 in^2

Big Fonts are NOT data Visualizations

64%

Parents look at the contents of their child's cell phone

Visual are perceived as more important



48%

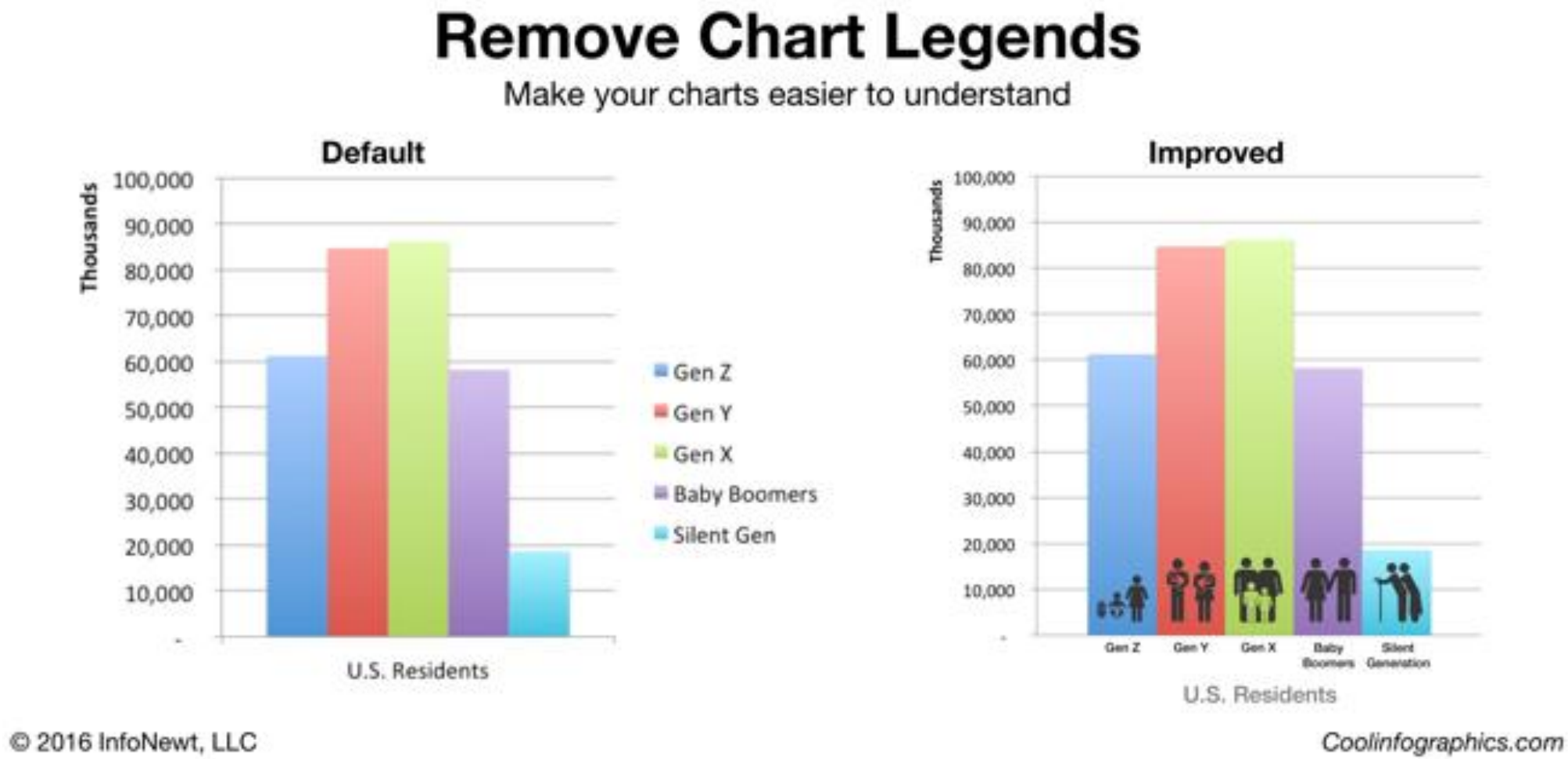
Parents use the phone
to monitor
their child's location

v. Simplify and Focus

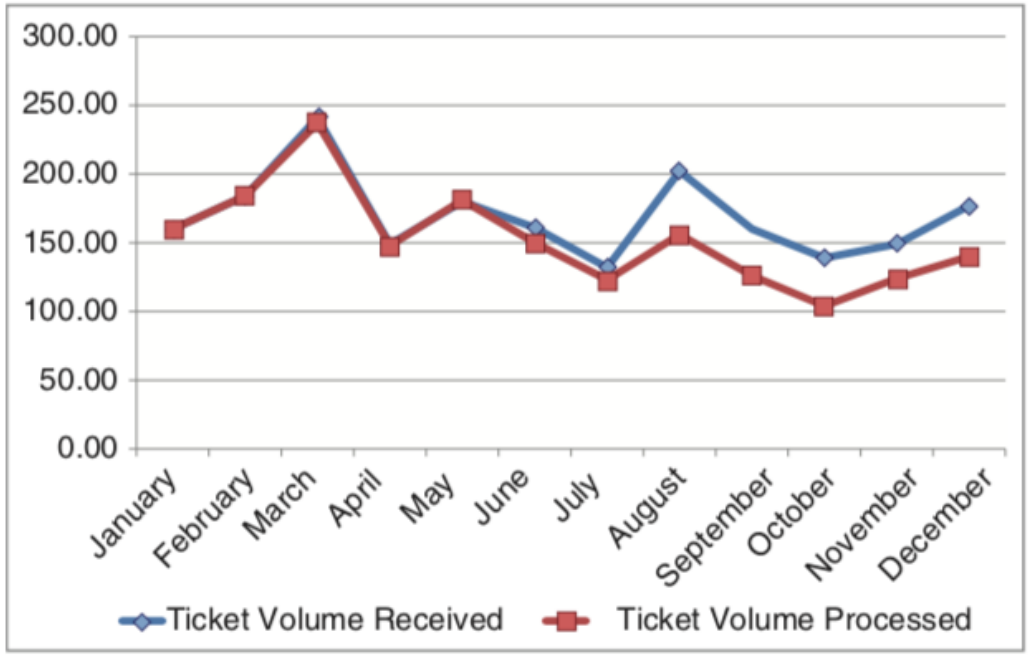
Remove unnecessary elements such as eliminate:

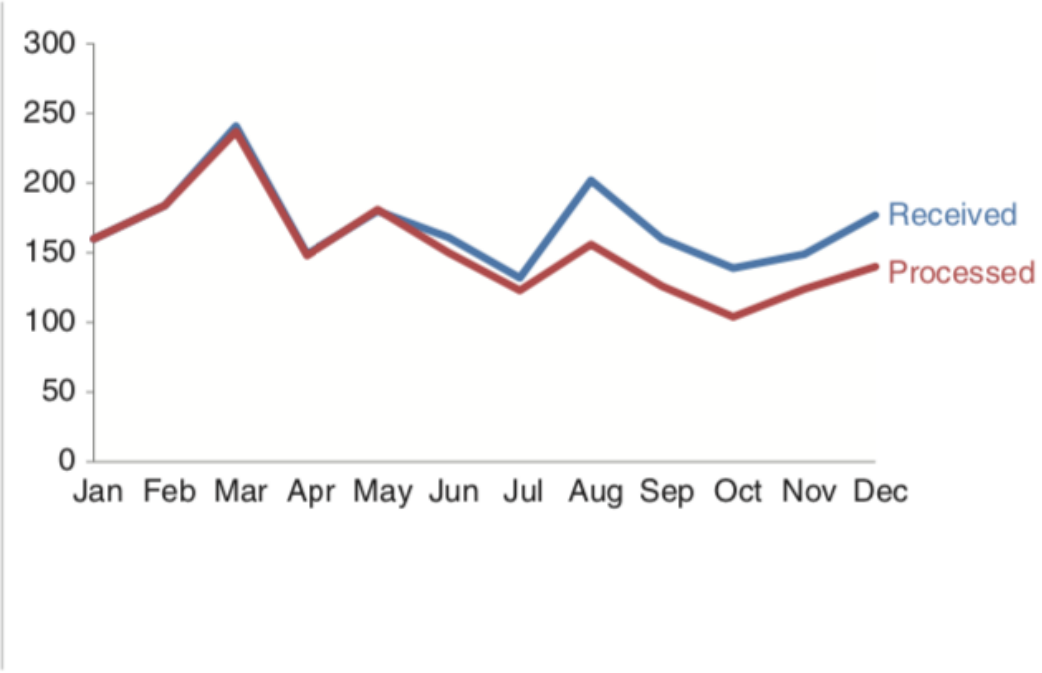
- 3D effects
- Redundant legends
- Unreadable fonts

Example


Remove Chart Legends
Make your charts easier to understand
Default
Thousands
100,000
90,000
80,000
70,000
60,000
50,000
40,000
30,000
20,000
10,000
-
Gen Z
Gen Y
Gen X
Baby Boomers
Silent Gen
U.S. Residents
Improved
Thousands
100,000
90,000
80,000
70,000
60,000
50,000
40,000
30,000
20,000
10,000
-
Gen Z
Gen Y
Gen X
Baby Boomers
Silent Generation
U.S. Residents
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Before and After

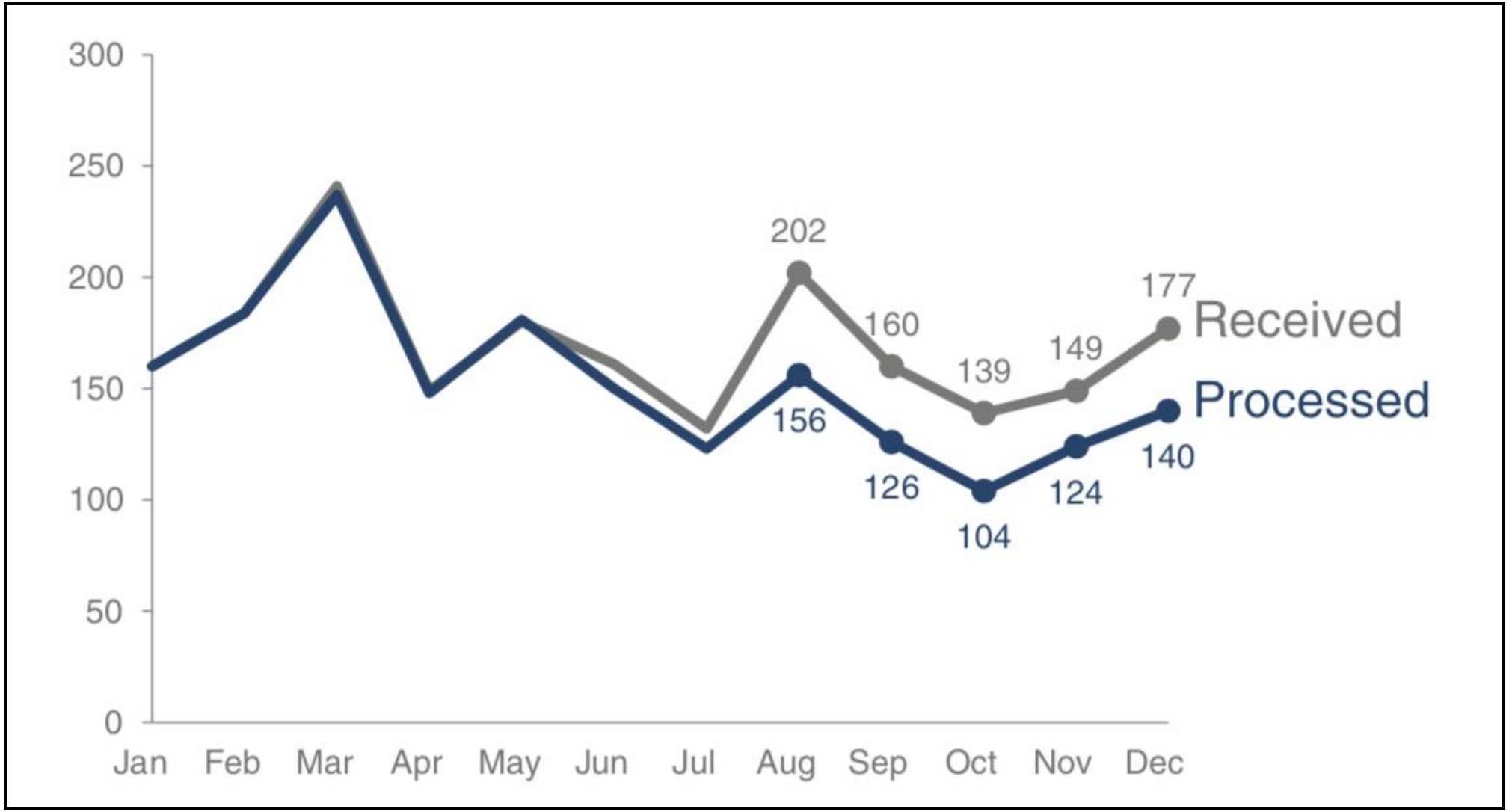




vi. Clear Labels & Annotations

- Use descriptive titles.
- Label axes and units.
- Add data labels or annotations where helpful.

Data labels used sparingly help draw attention

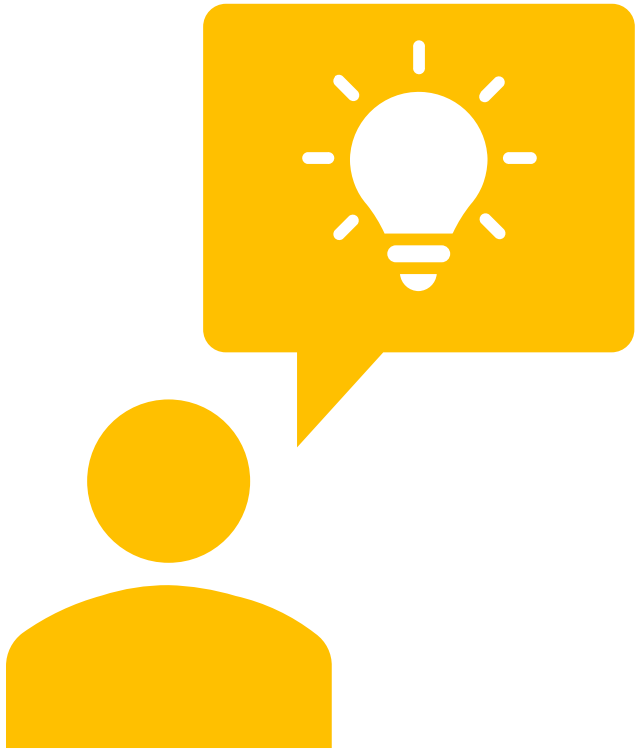




CLASS ACTIVITY

1. When might a line chart be more useful than a bar chart?
2. Why is a pie chart not ideal when comparing many categories?
3. Have you seen a chart that misrepresented data? Why?

KNOWLEDGE CHECK



THANK YOU



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